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Searle on Strong Artificial Intelligence and the Systems Reply

John Searle argues against what he labels as "strong AI (artificial intelligence)" in his 1980 paper "Minds, Brains, and Computers." The conception of computers has opened many new schools of thought in the pursuit of truth about the human brain and mind. Digital computers have challenged how we think about our own brain's process of performing mental tasks, such as understanding. In his paper, Searle acknowledges the ability of digital computers to complete tasks similar or identical to those done by human minds through programs and algorithms, but he challenges whether these computers truly understand the task they are computing. I seek to reconstruct Searle's argument through the premises that support it, and then elaborate further on what I suppose to be the strongest objection to his claim. First, I will define what I believe to be the most important term in Searle's paper: understanding. Then, I will iterate over his definitions of "weak" and "strong AI." Next, I will reconstruct Searle's argument against the possibility of a thinking digital computer by explaining the premises it is built on and what Searle concludes from these premises. Finally, I will end by explaining the Systems Reply to the Chinese Room analogy and explore how this objection finds some success in challenging the very definition of artificial intelligence.

Searle recognizes the term “understanding” to mean both the possession of mental states and the truth of these states. Searle is only concerned with the former definition; the validity of the states makes no difference in his case. Searle explains that we may say that certain objects understand things; that the calculator knows how to add, or the thermometer perceives changes in temperature. However, it is necessary to recognize that we extend our intentionality to these artifacts through our purposes for them. Furthermore, this is not the type of understanding that Searle is calling into question, but rather that the kind of cognition in computers is the same kind of cognition of the human mind. And based on that claim, just as he would say with the calculator and the thermometer, the understanding of the computer is nothing.

Searle differentiates between what he refers to as “weak AI” and “strong AI” within digital computers. He defines weak AI as the simulation of mental abilities, whereas strong AI really has mentality and mental states. Searle is not opposed to the existence of weak AI; he concedes it is a powerful tool and has no qualms with claiming that it is only a simulation. Searle is, however, starkly opposed to the existence of strong AI: artificial intelligence that has mental states, cognitive abilities, and is seen to be equivalent to the human mind. Searle devotes his paper to denying its current and future existence within the constraints of a digital computer.

To make his claim that digital AI does not function equally to the human mind, Searle establishes a basis. First, he proposes that digital computers run on syntax. Similar to how grammar influences the arrangement of words in language, syntax is the computer’s manipulation of symbols to compute outputs. Next, Searle contrasts the syntactic computer with the semantic human mind. He asserts that human minds are able to assign meaning to symbols beyond a syntax. Where a computer strictly interprets a symbol, humans interpret meaning, and this interpretation of meaning is at the core of understanding. Searle’s final premise uses the

analogy of the Chinese Room to illustrate that digital computers do not have the ability to assign meaning to symbols, and therefore the privilege of understanding is restricted to only human minds, or more specifically restricted from digital computers as Searle never rules out the possibility of a new kind of thinking machine.

Searle first establishes that digital computers are syntactic. This premise lies within the very nomenclature of the object. The term digital implies that the computers function on two commands, such as off/on or 0/1. The commands themselves do not matter; they could be indecipherable (to us) squiggles, but so long as they are symbols recognized by the computer, they serve the same function. The implementation of syntax is the implementation of a set of instructions given by the programmer that the computer then uses to interpret and manipulate the aforementioned symbols and turn inputs into outputs. Searle therefore claims that because digital computers are programmed only to interpret and manipulate symbols, they cannot do anything beyond that. He posits that the inputs and outputs of a computer contain no meaning to it, so if asked to do an arithmetic problem such as "2+3," the digital computer would answer "5" with no understanding of the answer itself. Where a human may associate "5" with years of age, objects present, or temperature in a room, the computer simply views 5 as the output that was generated when it followed instructions and manipulated symbols in the correct manner. The computer does not even know what "5" is, does not understand cardinality, or number theory, or arithmetic, only that "5" is the output.

Searle then establishes that human minds use semantics. The human brain developed in such a way that it excels at making connections. The arising abilities to categorize and associate stem from the exceptional capability of the brain to make connections due to wiring of neurons and neural pathways. Semantics is the way that human minds make associations and connections

to give meaning to symbols. To the human mind, symbols are more than a step of interpretation in a process of instructions, but rather contain deeper meanings within themselves. Searle also highlights that brains have causal powers, which cause minds, as well as intentionality, the power of minds to be about something such as beliefs, perceptions and desires. In order for computers to function cognitively equally to human minds, they must have both of these properties. Determining the presence of causal powers in hardware can be done by analyzing the intentionality of computers, which has not been observed to be present. Therefore, computers lack these two properties.

As mentioned when defining the usage of the term "understanding" in this paper, understanding applies to the argument of whether a digital computer functions equally to a human mind. I concede that a computer understands a program and syntax in the way that it is able to follow the instructions and operate on them, but this is not the definition of understanding which Searle seeks to establish within minds and exclude from digital computers. Through Searle's definition based on the argument he is conducting, understanding can only be found in semantic cognition. An example of this difference is in a chess-playing computer. Although the chess-playing computer can play chess better than a human player, the computer is not aware it is playing chess. It does not understand that it is competing in a game, but rather only knows what move to play next based on its algorithms which stem from its interpretation of symbols. The human, on the other hand, knows they are playing chess, knows they are at risk of losing the game, and are playing because it is fun to them. The human truly understands chess, where the computer only understands the syntactic manipulation of the symbols that eventually reach them through layers of simplification.

Searle's final premise is that digital computers cannot give meaning to symbols. Computers have no way to develop semantics. As Searle establishes prior, only semantics can lead to understanding, and in the employment of the Chinese Room analogy, Searle displays how syntax can never lead to semantics. The analogy is as follows: Suppose AI has progressed so far that there is a computer that behaves as if it understands Chinese. When given Chinese characters as input, it gives Chinese characters as output. It does this in such a convincing manner that it passes the Turing test; it convinces a human native Chinese speaker that it is also a human native Chinese speaker by successfully imitating one. Searle questions if the machine literally understands Chinese or if it merely simulates an understanding of Chinese: Strong AI versus weak AI. Searle then supposes a new scenario. Rather than a computer, he himself is in a closed room along with a database of filing cabinets composed of how to respond to any Chinese questions that he may be given. When he receives Chinese characters as input, he is able to follow the instructions in the database and respond with adequate Chinese characters as output despite not knowing any Chinese. His Chinese response is good enough to pass the Turing test. Searle claims that there is no essential difference between the computer's manipulation of the program and his own manual manipulation of the program, as both resulted in intelligent conversation. He then claims that just as he has no understanding of Chinese, it therefore follows that neither does the computer. Although the computer is able to create meaning, it itself is unable to give meaning to the Chinese characters, just as Searle is unable to give them meaning. Semantics, therefore, cannot arise from syntax alone.

Searle goes on to conclude that without understanding, the computer does not think, and because it does not think, the computer does not have a mind and therefore the strong AI hypothesis is false. Digital computers do not have minds and cannot create minds. Searle is

careful to limit this strictly to digital computers, and does not waive the possibility of the creation of alternative thinking machines. He even regards the brain itself as a thinking machine. Searle posits that so long as a machine fulfills the requirements of causal powers and intentionality, it can think and understand. However, so far only the brain is capable of fulfilling these functions, and so in a roundabout way Searle is saying that so far only brains can think.

The strongest objection to Searle's argument targets his Chinese Room analogy and how he interprets it. The Systems reply attempts to resolve the question of the location of the mind within the room that understands Chinese. The basic objection is that "the system understands Chinese." The combination of the person, the database, and the program form a system that understands Chinese. As Searle explains, understanding is not being ascribed to the mere individual, but to the system as a whole of which they are a part. Searle demands that an explanation must be made for how this system of objects has developed a mind. He responds by simplifying the situation by supposing that the person in the room becomes so proficient that they no longer consult the database and have the entire memorized program in their head. The person has become the system yet still does not understand Chinese. Responses to this simplification argue that the person now has two minds in his head; that if a mind is simply a way to process information, whether in a semantic or syntactic way, the person now has both: their semantic native language and their acquired syntactic Chinese. Searle regards this response as circular, as the question itself is if the mind is simply an information processor, and this response begs the question. Another version of the Systems reply is the virtual mind variation, where the mind in the system exists virtually within it just as files on a computer exist virtually. Searle responds by claiming that just as no one worries about the destructiveness of a virtual fire, the virtual mind is at best a simulation. However, for some things simulation is just as good as

the authentic thing. No one complains that a virtual calculator is not as good as a real one because the physical attributes do not matter; it is multiply realizable. The question this poses is whether the mind is essentially information, like the calculator, or something more, like the fire, that cannot be simulated. Perhaps AI of a certain caliber should be referred to in a different manner, such as synthetic intelligence.